

# \* List of Formulas

## \* Unit-1

### 1) Stress ( $\sigma$ )

(i) longitudinal stress ( $\sigma_l$ ) =  $\frac{F}{A}$

(ii) Volume stress ( $\sigma_v$ ) =  $\frac{F}{A} = \frac{PA}{A}$

$$\sigma_v = P$$

(iii) Shear stress =  $\frac{F_t}{A}$

→ Unit -  $N/m^2$

### 2) Strain ( $\epsilon$ )

(i) longitudinal strain ( $\epsilon_l$ ) =  $\frac{\Delta l}{L}$

(ii) Volume strain ( $\epsilon_v$ ) =  $\frac{\Delta V}{V}$

(iii) Shear strain ( $\epsilon_s$ ) =  $\frac{\Delta x}{L}$

3) Hook's law

$$\text{Modulus of Elasticity (E)} = \frac{\sigma}{\epsilon}$$

(i) Young's modulus (Y)

$$Y = \frac{mgL}{\pi r^2 \Delta L}$$

(ii) Bulk Modulus (K)

$$K = - \frac{PV}{\Delta V}$$

(iii) Modulus of rigidity ( $\eta$ )

$$\eta = \frac{F_t \cdot l}{\Delta x \cdot A}$$

4) Poisson's Ratio ( $\nu$ )

$$\nu = \frac{\Delta d / d}{\Delta l / l}$$

4) Young's Modulus - Non-uniform bending

$$Y = \frac{Wl^3}{4ybd^3}$$

5) Young's Modulus - Uniform bending.

$$Y = \frac{W \times l^2}{8y I_g} \quad \therefore I_g = \frac{bd^3}{12}$$

6) Twisting Couple ( $\tau$ )

$$\tau = \frac{\pi \eta \theta R^4}{2L}$$

7) Torsional Rigidity ( $G$ )

$$G = \frac{\pi \eta R^4}{2L}$$

8) Time period ( $T$ )

$$T = 2\pi \sqrt{\frac{I}{G}}$$



## Unit-2

### Waves and Motion

\* For S.H.M.

1. Displacement

$$x = A \sin(\omega t + \phi)$$

2. Velocity

$$V = \omega A \cos(\omega t + \phi) \Rightarrow V_{\max} = \omega A$$

3. Acceleration

$$a = -\omega^2 A \sin(\omega t + \phi) \\ = -\omega^2 x$$

$$\text{where, } \omega = \frac{2\pi}{T} \quad \text{or} \quad \omega = \sqrt{\frac{K}{m}}$$

## Unit-2

1) Frequency of vibrations of crystal

$$f = \frac{p}{2L} \sqrt{\frac{E}{\rho}}$$

$p = 1, 2, 3 \dots$  for Fundamental, first overtone, second overtone...

2) Sabine's Formula

$$T = \frac{0.167V}{\sum aS}$$

where,  $V$  = Volume of hall  
 $a$  = absorption coefficient  
 $S$  = Surface area.

$$\rightarrow \text{Average absorbing Power} = \frac{\sum aS}{S}$$

3) Intensity level ( $I_L$ )

$$I_L = 10 \log \left( \frac{I}{I_0} \right) \quad I_0 = 10^{-12} \text{ Watt/m}^2$$

## Unit-4

1) Fringe width  $\beta = \frac{\lambda D}{d}$

$D$  = distance bet<sup>n</sup> screen & slit.

$d$  = distance bet<sup>n</sup> two slits.

2) Newton's Ring

i) Radius of curvature

$$R = \frac{D_{n+p}^2 - D_n^2}{2p\lambda}$$

ii) Diameter of Dark ring

$$D_n^2 = 4n\lambda R$$

Condition for minima

$$2t = n\lambda$$

3) Resolving Power (R.P) =  $\frac{\lambda}{d\lambda}$

where,  $\lambda = \frac{\lambda_1 + \lambda_2}{2}$

$$d\lambda = \lambda_2 - \lambda_1$$



3) Thin Film  
 → Condition for minima (dark band):

$$2\mu t \cos r = n\lambda$$

4) Snell's law

$$\mu = \frac{\sin i}{\sin r}$$

5) Anti-Reflecting Coating

$$t_{\min} = \frac{\lambda}{4\mu_f}$$

## Unit-3

$$1) E = h\nu = \frac{hc}{\lambda}$$

$$2) E = \frac{n^2 h^2}{8mL^2} \quad (n=1, 2, 3, \dots)$$

3) De-Broglie Wavelength.

$$(i) \lambda = \frac{h}{p} = \frac{h}{mv}$$

$$(ii) \lambda = \frac{h}{\sqrt{2mE}}$$

$$(iii) \lambda = \frac{h}{\sqrt{2meV}}$$

$$(iv) \lambda = \frac{h}{\sqrt{3mkT}}$$

4) Heisenberg's Uncertainty Principle

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$



## Unit-5 LASERS.

1.  $E = \frac{hc}{\lambda}$

$$E_t = N \times E$$

where  $N$  = No. of atoms.

2.  $\frac{N_2}{N_1} = e^{-\frac{E}{k_B T}}$

$$1 \text{ eV} = 1.6 \times 10^{-19} \text{ Joule.}$$

(i)  $\text{eV} \rightarrow \text{J} : \text{Multiply by } 1.6 \times 10^{-19}$

(ii)  $\text{J} \rightarrow \text{eV} : \text{Divide by } 1.6 \times 10^{-19}$

## Unit-5.

$$1) E = \frac{hc}{\lambda}$$

$$2) \frac{N_2}{N_1} = e^{-h\nu/kT}$$
$$= e^{-hc/\lambda kT}$$

## 3) Unit-6.

1) Critical temperature

$$T_c M^\alpha = \text{const}$$

$$T_c \propto \frac{1}{\sqrt{M}}$$

$$T_{c1} M_1^\alpha = T_{c2} M_2^\alpha$$

(ii) Critical Magnetic field

$$H_c = H_{c0} \left[ 1 - \left( \frac{T}{T_c} \right)^2 \right]$$

(iii) Critical Current.

$$I_c = 2\pi r H_c$$